Plain Language Translation Datasets

This document describes the datasets included in `merged_plain_language_dataset.csv`.

CELLS

Source identifier: `CELLS`

Biomedical Lay Summarization dataset containing paired abstracts and lay summaries from scientific articles. The dataset focuses on transforming technical biomedical abstracts into accessible summaries written for general audiences. It is the largest dataset in this collection with over 62,000 pairs, making it valuable for training text simplification models. The data includes predefined train/validation/test splits.

Reference: Guo, Y., Qiu, W., Leroy, G., Wang, S., & Cohen, T. (2024). Retrieval augmentation of large language models for lay language generation. *Journal of Biomedical Informatics*, 149, 104596. (Dataset originally introduced in arXiv:2211.03818, 2022)

Source: https://github.com/LinguisticAnomalies/pls_retrieval

Cochrane

Source identifier: `Cochrane`

Medical systematic review simplification dataset scraped from the Cochrane Library. Each entry pairs a technical abstract from a Cochrane systematic review with its corresponding Plain Language Summary (PLS) written for patients and the public. The dataset includes both "long" format (single paragraph) and "sectioned" format (multiple headed sections) plain language summaries. This collection uses the raw, unfiltered JSON data to preserve maximum coverage.

Reference: Devaraj, A., Marshall, I., Wallace, B., & Li, J. J. (2021). Paragraph-level simplification of medical texts. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 4972-4984.

Source: https://github.com/AshOlogn/Paragraph-level-Simplification-of-Medical-Texts

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PLABA

Source identifier: `PLABA`

Plain Language Adaptation of Biomedical Abstracts dataset containing sentence-level simplifications of PubMed abstracts. Each abstract sentence was adapted by human annotators to be more accessible to lay readers. Approximately 22% of the abstracts have simplifications from two different annotators, with both versions included as separate rows to capture annotation variation. The dataset was created for the TREC 2024 PLABA track focused on biomedical text simplification.

Reference: Attal, K., Ondov, B., & Demner-Fushman, D. (2023). A dataset for plain language adaptation of biomedical abstracts. *Scientific Data*, 10, 8.

Source: https://osf.io/rnpmf/ (TREC PLABA: https://trec-plaba.github.io/)

Med-EASi

Source identifier: `Med-EASi`

Medical text simplification dataset with expert-written sentences paired with simplified versions. The dataset includes detailed edit annotations showing the specific transformations applied during simplification (lexical substitution, sentence splitting, elaboration, etc.). It provides a fine-grained view of the simplification process beyond just source-target pairs. The data is hosted on Hugging Face and includes train/validation/test splits.

Reference: Basu, C., Vasu, R., Yasunaga, M., & Yang, Q. (2023). Med-EASi: Finely annotated dataset and models for controllable simplification of medical texts. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(12), 14093-14101.

Source: https://huggingface.co/datasets/cbasu/Med-EASi

MedLane

Source identifier: `MedLane`

Clinical notes simplification dataset focusing on transforming clinical language into patient-friendly text. The full dataset requires MIMIC-III access due to containing clinical notes from real patient records. This collection includes only the publicly available sample data (400 pairs) demonstrating the simplification approach. The samples show third-person to second-person transformations along with medical terminology simplification.

Reference: Luo, J., Lin, J., Lin, C., Xiao, C., Gui, X., & Ma, F. (2022). Benchmarking automated clinical language simplification: Dataset, algorithm, and evaluation. *Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022)*, 3546-3559.

Source: https://github.com/machinelearning4health/MedLane

CARES

Source identifier: `CARES`

Clinical trial plain language summary dataset pairing study protocols/synopses with patient-facing summaries. Each entry represents a complete clinical trial document (NCT-identified) with its corresponding plain language summary organized by sections (why was this study done, what happened, results, etc.). The source documents are unstructured clinical study reports extracted from PDF documents. This dataset captures document-level simplification rather than sentence-level.

Reference: Giannouris, P., Myridis, T., Passali, T., & Tsoumakas, G. (2024). Plain language summarization of clinical trials. *Proceedings of the Workshop on Evaluating Text Difficulty in a Multilingual Context (DeTermIt!) at LREC-COLING 2024*, 60-67.

Source: https://github.com/PolydorosG/CARES

PERCS

Source identifier: `PERCS`

PubMed Expert-to-Layman Rewriting Corpus for Simplification containing multi-persona summaries of biomedical abstracts. Each abstract was summarized by four different personas: layman, pre-med student, researcher, and expert. This collection pairs the expert summaries with layman summaries, capturing the transformation from technical expert language to accessible lay language. The dataset includes 500 abstracts with predefined train/test splits.

Reference: Salvi, R. C., et al. (2024). PERCS: Persona-guided controllable biomedical summarization dataset. *arXiv preprint arXiv:2512.03340*.

Source: https://osf.io/umbj7/

PlainQAFact

Source identifier: `PlainQAFact`

Question-answering based simplification dataset pairing biomedical abstracts with factually accurate simplified sentences. The dataset distinguishes between factual and non-factual simplifications, with this collection using only the factual target sentences. Each row maps an original abstract to a simplified sentence that accurately conveys the key information. The focus on factual accuracy makes this dataset valuable for training models that maintain information fidelity during simplification.

Reference: You, Z., Guo, Y., Leroy, G., Xu, D., Shang, J., & Cohen, T. (2025). PlainQAFact: Retrieval-augmented factual consistency evaluation metric for biomedical plain language summarization. *arXiv preprint arXiv:2503.08890*.

Source: https://github.com/zhiwenyou103/PlainQAFact

Summary Table

Source Identifier	Rows	Level	Splits Available
CELLS	62,886	Abstract	train/val/test

Source Identifier	Rows	Level	Splits Available
PLABA	9,085	Sentence	none
Cochrane	7,816	Paragraph	none
PlainQAFact	2,740	Sentence	none
Med-EASi	1,893	Sentence	train/val/test
PERCS	500	Abstract	train/test
MedLane	400	Sentence	train/test
CARES	169	Document	train/test
Total	**85,489**		